Chapter-1

Operating system concepts

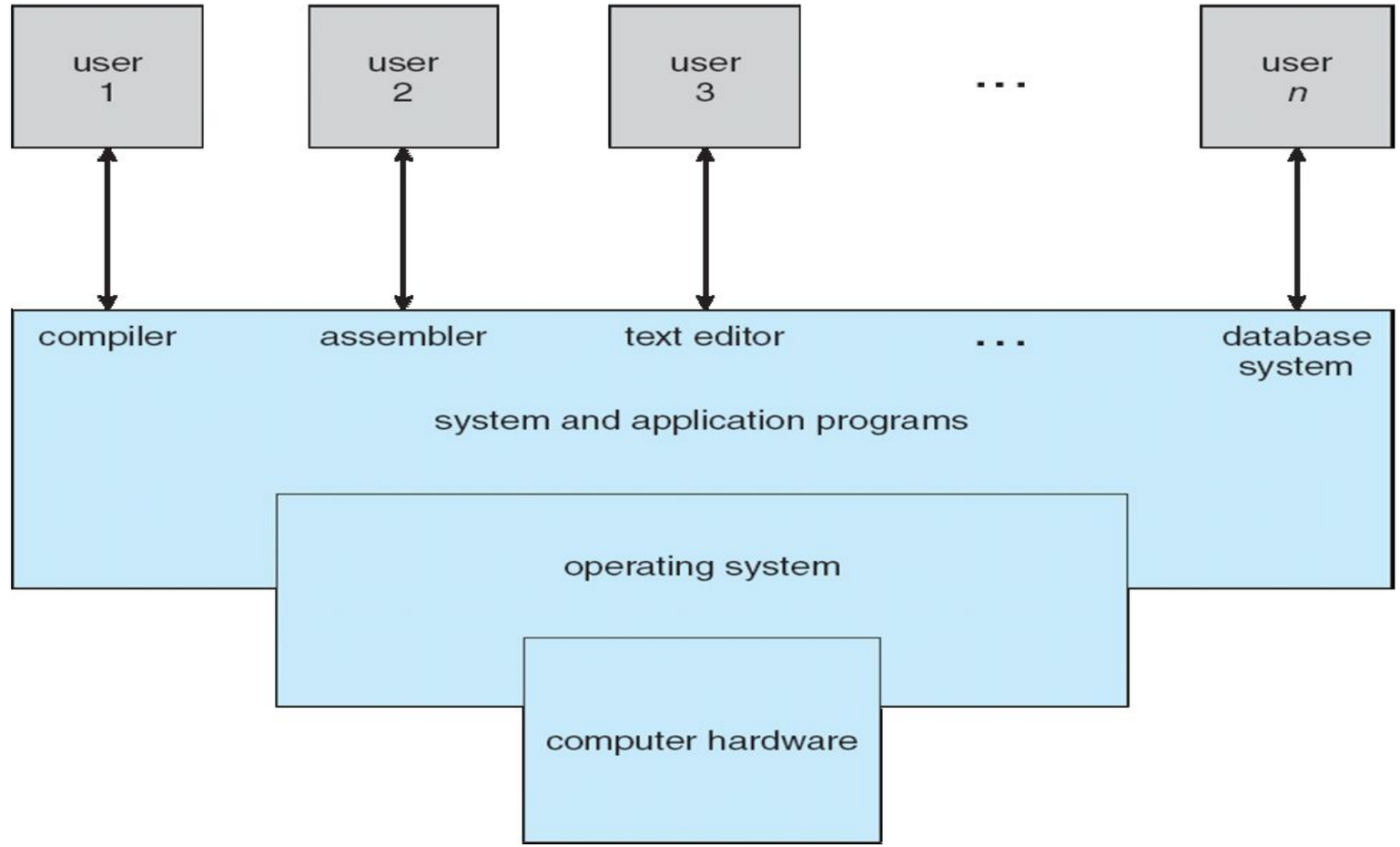
What is an Operating System?

- Computer System = Hardware + Software
- Software = Application Software + System Software(OS)
- An Operating System is a system Software that acts as an intermediary/interface between a user of a computer and the computer hardware.
- Operating system goals:
  - Execute user programs and make solving user problems easier
  - Make the computer system convenient to use
  - Use the computer hardware in an efficient manner

Computer System Structure

- Computer system can be divided into four components:
  - **Hardware** – provides basic computing resources
    - CPU, memory, I/O devices
  - **Operating system**
    - Controls and coordinates use of hardware among various applications and users
  - **Application programs** – define the ways in which the system resources are used to solve the computing problems of the users
    - Word processors, compilers, web browsers, database systems, video games
  - **- Users**
    - People, machines, other computers

Four Components of a Computer System



- Operating systems can be explored from two viewpoints:

1. User view
2. System view

Evolution (types) of operating systems

- What is Job?

✔ Unit of work that a computer operator gives to the operating system.

✔ For example, a job could be the running of an application program such as a weekly payroll program.

- **Sequential processing-**

✔ A sequential operating system (or more accurately, sequential processing/access) executes tasks or processes data one after another, in a strict, ordered sequence, where one step must finish before the next begins, like reading a book line by line.

✔ **Key Characteristics-**

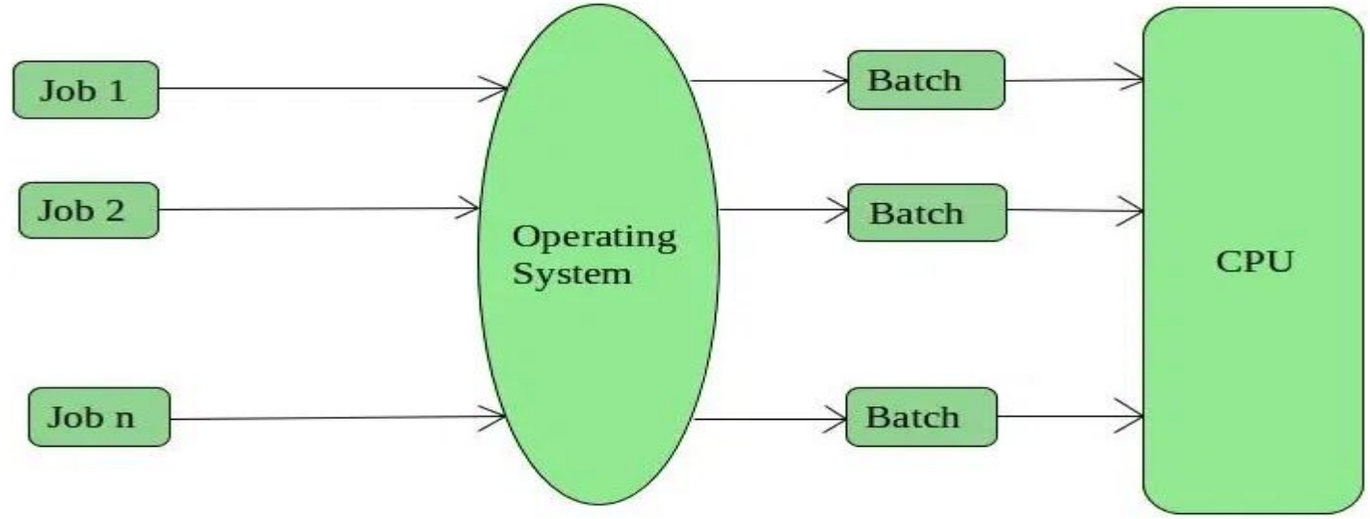
1. **One Task at a Time**: Only one instruction or job runs at any given moment.
2. **Simplicity**: Easier to implement and understand.
3. **Inefficient for Random Access**: Slow if you need data from the middle or end, requiring traversal from the start.

- **Batch processing**

✔ A Batch Operating System is designed to handle large groups of similar jobs efficiently.

✔ It does not interact with the computer directly but instead processes jobs that are grouped by an operator.

✔ These jobs are queued and executed one after the other, without user interaction during the process.



Advantages of Batch Operating System

1. **Minimal Idle Time**: The system minimizes idle time by processing jobs in a continuous sequence without human intervention.
2. **Handling Repetitive Tasks**: Ideal for managing large, repetitive tasks, such as payroll and billing, with minimal effort.
3. **Improved Throughput**: Batch systems can handle high volumes of jobs at once, improving overall system throughput.

Disadvantages of Batch Operating System

1. **Inefficient CPU Utilization**: When a job is waiting for input/output (I/O), the CPU remains idle, leading to poor utilization of resources.
2. **Increased Response Time**: The time between job submission and output can be high as all jobs are processed sequentially.
3. **Lack of Real-Time Feedback**: Users cannot interact with the system in real-time, making it less suitable for interactive tasks.

Examples:

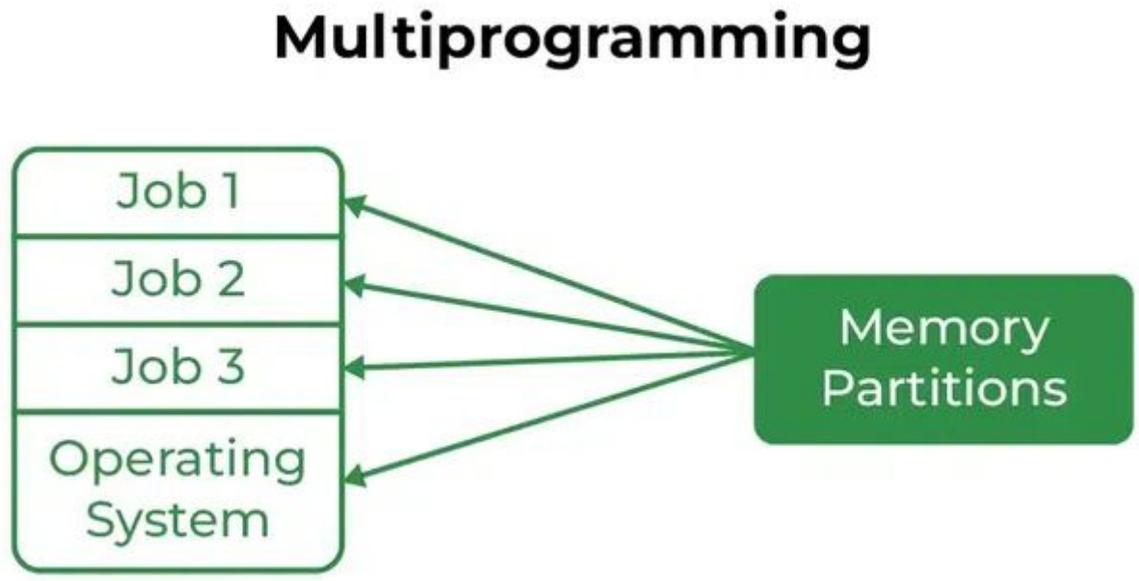
Payroll Systems

Bank Statements

- **Multi-Programming Operating System**

✔ In a Multi-Programming Operating System, multiple programs run in memory at the same time.

✔ The CPU switches between programs, utilizing its resources more effectively and improving overall system performance.



- **Advantages**

1. **Better CPU Utilization**: CPU stays busy by switching to another job during I/O wait.
2. **Improved Throughput**: Multiple jobs run concurrently, increasing work done per unit time.
3. **Efficient Resource Use**: CPU, memory, and I/O devices are shared effectively among processes.

Disadvantages

1. **Complex Design**: Requires advanced memory management and CPU scheduling.
2. **Security Issues**: More programs in memory increase chances of unauthorized access.
3. **High Memory Requirement**: Needs larger RAM to run multiple programs together.

Examples-

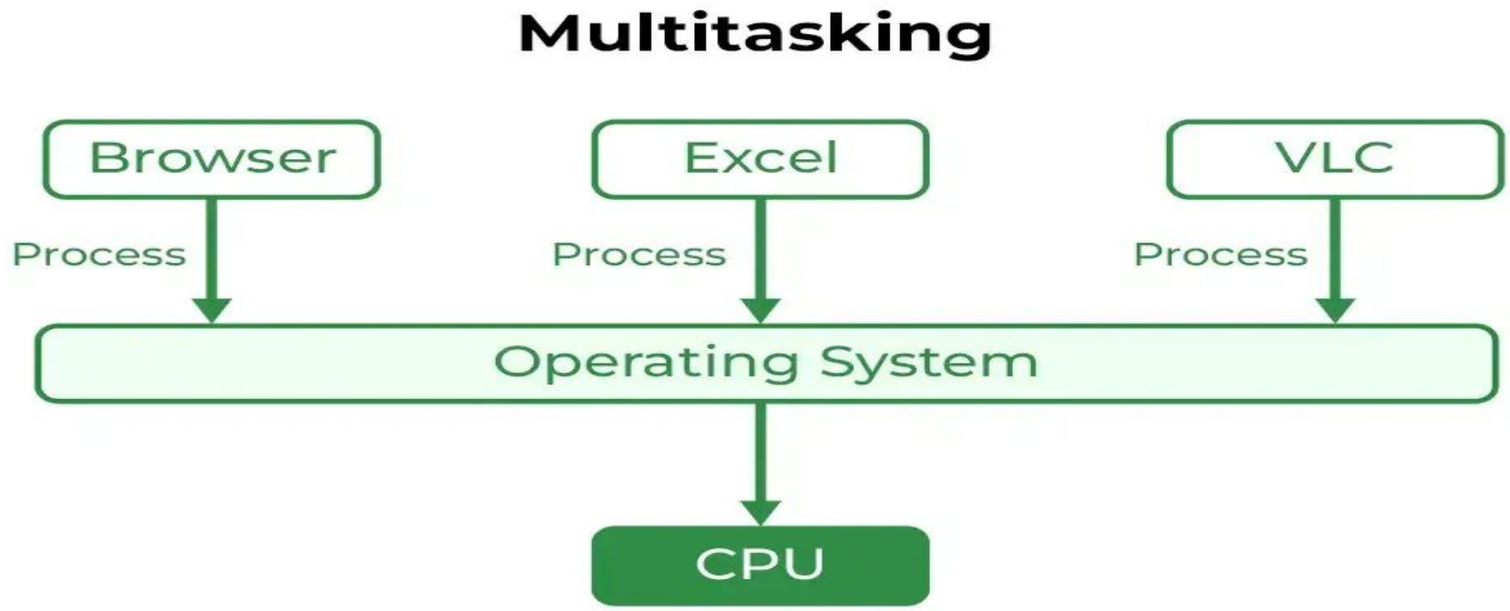
Banking systems
Railway servers
Billing machines

- **Multi-tasking/Time-sharing Operating systems-**

✔ Multitasking OS is a type of multiprogramming system where each process runs in a round-robin manner.

✔ Every task gets a fixed time slice called a quantum.

✔ After the quantum ends, the OS switches to the next task, allowing multiple tasks—whether from one user or many—to run smoothly on a single system.



- **Advantages**

1. **Equal CPU Access**: Each task gets a fair share of CPU time.
2. **Reduced Software Duplication**: Many users can run the same software without needing separate copies.
3. **Low CPU Idle Time**: Efficient scheduling keeps the CPU busy.

- **Disadvantages**

1. **Lower Reliability:** System failures affect all users.
2. **Security Concerns**: Multiple users increase risks to data integrity and privacy.
3. **Communication Issues:** Data sharing between users can cause conflicts.

- **Real-Time Operating System**

✔ These types of OSs serve real-time systems.

✔ The time interval required to process and respond to inputs is very small. This time interval is called response time.

✔ Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

- **Types of Real-Time Operating Systems**

1. **Hard Real-Time Operating System:**

✔ Used where strict timing is essential and any delay is unacceptable, such as airbags or automatic parachutes.

✔ These systems avoid virtual memory to ensure immediate response.

2. **Soft Real-Time Operating System**:

✔ Used where timing is important but minor delays are acceptable.

✔ These systems aim to give quick and predictable responses but do not require perfect accuracy.

✔ They are commonly used in multimedia applications, gaming, video streaming, and other interactive tasks.

- **Advantages**

1. **Maximum Consumption**: Maximum utilization of devices and systems, thus more output from all the resources.
2. **Error-Free**: These types of systems are error-free.
3. **Memory Allocation**: Memory allocation is best managed in these types of systems.

- **Disadvantages**

1. **Limited Tasks:** Very few tasks run at the same time and their concentration is very less on a few applications to avoid errors.
2. **Complex Algorithms**: The algorithms are very complex and difficult for the designer to write.
3. **Thread Priority**: It is not good to set thread priority, as these systems are much less prone to switching tasks.

Examples:

Scientific experiments
Medical imaging systems
Robots

- **Multithreading OS –**

✔ Multithreading is a technique where a process is divided into smaller execution units called threads that run concurrently.

✔ A thread is also called a lightweight process.

✔ Concurrency or Parallelism within a process is achieved by dividing a process into multiple threads.

✔ Multithreading improves system performance and responsiveness by allowing multiple threads to share CPU, memory and I/O resources of a single process.

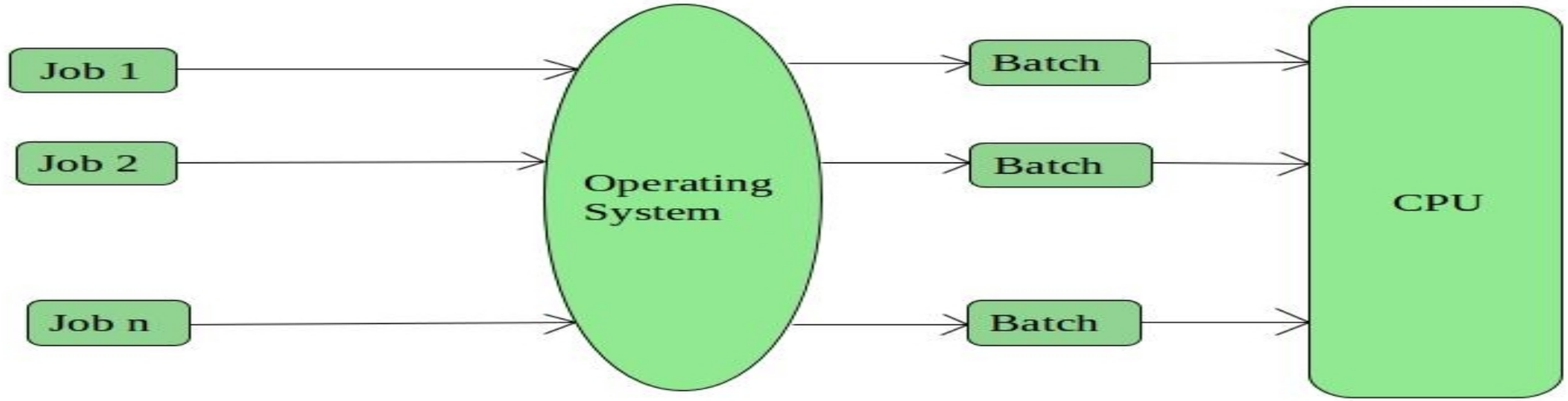
Example: In a browser, each tab can be a thread. In MS Word, one thread formats text while another processes inputs.

Types of Operating Systems

1. Batch Operating System
2. Multiprogramming Operating System
3. Time-Sharing OS
4. Multiprocessing OS
5. Real Time OS

1. Batch Operating System

- The users of this type of operating system does not interact with the computer directly.
- Each user prepares his job on an off-line device like punch cards and submits it to the computer operator
- There is an operator which takes similar jobs having the same requirement and group them into batches.



Batch Operating System cont..

Advantages of Batch Operating System:

- Processors of the batch systems know how long the job would be when it is in queue
- Multiple users can share the batch systems
- The idle time for the batch system is very less
- It is easy to manage large work repeatedly in batch systems

Disadvantages of Batch Operating System:

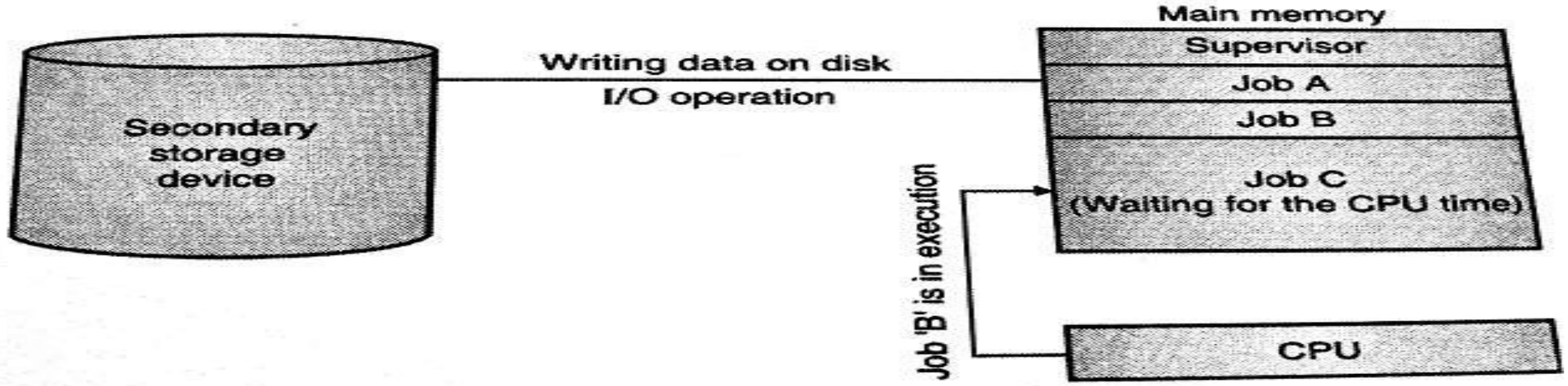
- The computer operators should be well known with batch systems
- Batch systems are hard to debug
- It is sometimes costly
- The other jobs will have to wait for an unknown time if any job fails

Examples of Batch based Operating System:

IBM's MVS

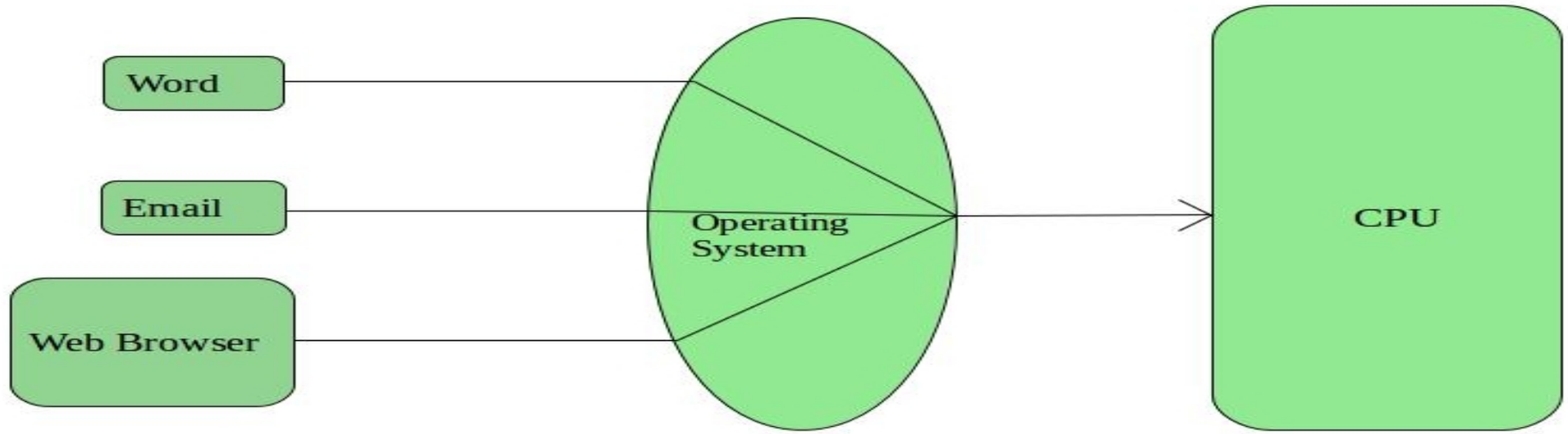
2. Multiprogramming Operating System:

- This type of OS is used to execute more than one jobs simultaneously by a single processor.
- It increases CPU utilization by organizing jobs so that the CPU always has one job to execute.
- Multiprogramming operating systems use the mechanism of job scheduling and CPU scheduling.



3. Time-Sharing Operating Systems

- Each task is given some time to execute so that all the tasks work smoothly.
- These systems are also known as **Multi-tasking Systems.**
- The task can be from a single user or different users also.
- The time that each task gets to execute is called quantum.
- After this time interval is over OS switches over to the next task.



3. Time-Sharing Operating Systems cont..

- **Advantages of Time-Sharing OS:**
  - Each task gets an equal opportunity
  - Fewer chances of duplication of software
  - CPU idle time can be reduced
- **Disadvantages of Time-Sharing OS:**
  - Reliability problem
  - One must have to take care of the security and integrity of user programs and data
  - Data communication problem
- **Examples of Time-Sharing Oss**

  Multics, Unix, etc.

4. Multiprocessor operating systems

- Multiprocessor operating systems are also known as parallel OS or tightly coupled OS.
- Such operating systems have more than one processor in close communication that sharing the computer bus, the clock and sometimes memory and peripheral devices.
- It executes multiple jobs at the same time and makes the processing faster.
- It supports large physical address space and larger virtual address space.
- If one processor fails then other processor should retrieve the interrupted process state so execution of process can continue.
- Inter-processes communication mechanism is provided and implemented in hardware.

7. Real-Time Operating System

- These types of OSs serve real-time systems.
- The time interval required to process and respond to inputs is very small.
- This time interval is called **response time**.
- **Real-time systems** are used when there are time requirements that are very strict like
  - missile systems,
  - air traffic control systems,
  - robots, etc.

• Operating System services

- There are various types of services provided by operating system.
- These services are utilized with help of system calls
- Following are types of services:

1. Program execution
2. i/o operation
3. File system operation
4. Communication
5. Error detection
6. Resource allocation
7. Accounting
8. Protection and security

- An Operating System provides services to both the users and to the programs.
- It provides programs an environment to execute.
- It provides users the services to execute the programs in a convenient manner.

- **System Call**

- A **system call** is the mechanism used by an application program to request service from the operating system.
- A system call is a **request** made by any arbitrary program to the operating system for performing tasks. System calls provide the interface between a process and the operating system itself.
- Application program is just a user-process. Due to security reasons , user applications are not given access to privileged resources(the ones controlled by OS).

- When they need to do any I/O or have some more memory or spawn a process or wait for signal/interrupt, it requests operating system to facilitate all these. This request is made through System Call.
- System calls are also called software-interrupts.

The main types of system calls are

1. **Process Control**: These types of system calls are used to control the processes.

Ex: End, abort, load, execute, create process, terminate process etc.

2. **File Management :**

These types of system calls are used to manage files.

Ex: Create file, delete file, open, close, read, write, get file attributes, set file attributes etc.

3. **Device Management :**

These types of system calls are used to manage devices.

Ex: Request device, release device, read, write, get device attributes

4. **Information maintenance:**

- get time or date, set time or date
- get system data, set system data
- get process, file, or device attributes
- set process, file, or device attributes

5.**Communications**:

- create, delete communication connection
- send, receive messages
- transfer status information
- attach or detach remote devices